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Shock absorber and bracket

Abstract

Provided are a shock absorber and a bracket which are improved in attachment rigidity. A cylindrical portion is shaped by press-forming first and second holes in sheet material (flat plate) and then curving the sheet material into a tube, in which a bridge portion is formed between the first and second holes. The bracket is thus improved to have higher formability as compared to conventional brackets which are not provided with a bridge portion. This improves the accuracy (cylindricity, in particular) of the cylindrical portion. It is then possible to improve the shock absorber in attachment rigidity against a lateral force to the anteroposterior direction of a vehicle in a situation where the bracket is used to attach the shock absorber to the knuckle.

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Background/Summary

TECHNICAL FIELD

(1) The invention relates to a shock absorber disposed between a vehicle body and a wheel and a bracket that is provided at an outer periphery of a shock absorber and coupled to a wheel-side knuckle.

BACKGROUND ART

(2) Patent Literature 1 discloses a bracket (hereinafter, referred to as a “conventional bracket”) including a tube-like portion (cylindrical portion) having a C-shaped cross section which encloses the outer periphery of an outer shell and has a split in a front portion, and a pair of attachment portions (extending portions) protruding radially outwards from respective circumferential ends of the tube-like portion, the tube-like portion being provided with a hole that is formed to avoid interference with a protruding portion (valve case).

CITATION LIST

Patent Literature

(3) PTL 1: Japanese Patent Application Publication (Kokai) No. 2018-25224

SUMMARY OF INVENTION

Technical Problem

(4) A conventional bracket includes a hole for allowing a protruding portion to extend therethrough. The hole is formed to extend from both side portions to a rear portion of the tube-like portion. The tube-like portion is shaped by press-forming a hole in sheet material (flat plate) and then curving the sheet material into a tube in which the hole is formed. However, it is difficult to shape the sheet material (flat plate) that is widely open at the center into a tube with high accuracy (cylindricity, in particular). The accuracy of the tube-like portion is related to shock absorber attachment rigidity against a lateral force to the anteroposterior direction of a vehicle in a situation where the shock absorber is attached to a knuckle through the bracket. An issue is therefore to improve attachment rigidity by improving the formability of the bracket.

(5) An object of the invention is to secure attachment rigidity of a shock absorber in which a hole for avoiding interference of a bracket with a protruding portion is formed and provide a bracket that is capable of securing attachment rigidity of a shock absorber.

Solution to Problem

(6) A shock absorber of the invention comprises a tube-like cylinder, a protruding portion that is so provided as to protrude in a radially outward direction of the cylinder, and a bracket that is provided at an outer periphery of the cylinder. The bracket comprises a cylindrical portion including a first end portion and a second end portion in a circumferential direction of the cylindrical portion, the cylindrical portion having a cross-section in an axial direction of the cylindrical portion, the cross-section being formed into a C-shape, the cylindrical portion covering

the outer periphery of the cylinder; a pair of extending portions respectively extending from the first end portion and the second end portion of the cylindrical portion in the radially outward direction of the cylindrical portion so as to face each other; a first hole that is provided in the cylindrical portion and allows the protruding portion to extend through the first hole in the radially outward direction of the cylindrical portion; a second hole that is provided in the cylindrical portion and disposed apart from the first hole; and a bridge portion that is formed between the first hole and the second hole and provided on an opposite side of an axial center of the cylindrical portion from the pair of extending portions.

(7) A bracket of the invention is a bracket that is provided to cover an outer periphery of a shock absorber disposed between a vehicle body and a wheel and coupled to a knuckle on the wheel side. The bracket comprises a cylindrical portion including a first end portion and a second end portion in a circumferential direction of the cylindrical portion, the cylindrical portion having a cross-section in an axial direction of the cylindrical portion, the cross-section being formed into a C-shape; a pair of extending portions respectively extending from the first end portion and the second end portion of the cylindrical portion in a radially outward direction of the cylindrical portion so as to face each other; a first hole that is provided in the cylindrical portion and extends through the cylindrical portion; a second hole that is provided apart from the first hole and extends through the cylindrical portion; a bridge portion that is formed between the first hole and the second hole on a circumferentially opposite side of the cylindrical portion from a center position of the cylindrical portion which is located between the first end portion and the second end portion; and reinforcing portions that are provided in the radially outward direction of the cylindrical portion and formed to extend from the cylindrical portion to the respective extending portions.

(8) One embodiment of the invention makes it possible to secure attachment rigidity of a shock absorber in which a hole for avoiding interference of a bracket with a protruding portion is formed.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view showing an attachment structure of a shock absorber according to an embodiment of the invention.

(2) FIG. 2 is a perspective view of the shock absorber of the embodiment as viewed from a first hole side.

(3) FIG. 3 is a perspective view of the shock absorber of the embodiment as viewed from a second hole side.

(4) FIG. 4 is a top view (conceptual diagram) of a bracket according to the embodiment.

(5) FIG. 5 is a left side view of the bracket according to the embodiment.

(6) FIG. 6 is a rear view of the bracket according to the embodiment.

DESCRIPTION OF EMBODIMENTS

(7) One embodiment of the invention will be explained with reference to the attached drawings.

(8) FIG. 1 is a perspective view showing a condition in which a shock absorber **1** of a suspension apparatus, not shown, is coupled to a wheel-side knuckle **10**. For the sake of convenience, an up-and-down direction in FIG. 1 will be referred to as “vertical direction” or “axial direction.” The shock absorber **1** is a damping force adjustable hydraulic shock absorber with a laterally-mounted control valve in which a valve device (only a valve case **6** of which is illustrated in FIG. 1) is laterally mounted on a lateral wall of a cylinder **2** (outer tube).

(9) As illustrated in FIGS. 1 to 3, the shock absorber **1** includes a piston rod **3** protruding outside the cylinder **2**. A spring bearing **4** and a bracket **5** are joined to an outer peripheral surface **2A** of the cylinder **2**. The spring bearing **4** receives a lower end of a coil spring, not shown. Linkage components and the like are connected to the bracket **5**. A valve case **6** (protruding portion) having

a cylindrical shape is provided to the lateral wall of the cylinder **2** to protrude in a radially outward direction of the cylinder **2**.

(10) The shock absorber **1** includes a bracket **11** that is provided to cover an outer periphery of the cylinder **2**. The shock absorber **1** has a similar structure to a conventional damping force adjustable hydraulic shock absorber with a laterally-mounted control valve, except for being provided with the bracket **11**. For the purpose of brevity of written description, the description will omit a detailed explanation on such a damping force adjustable hydraulic shock absorber with a laterally-mounted control valve.

(11) Referring to FIGS. **4** to **6**, the bracket **11** includes a cylindrical portion **21**. A cross-section in an axial direction (cross-section taken along a plane perpendicular to an axial line of the cylinder **2**) of the cylindrical portion **21** is formed into a C-shape. The bracket **11** includes a pair of extending portion **31** and extending portion **41** which extend from a first end portion **22** and a second end portion **23** at respective circumferential ends of the cylindrical portion **21** radially outwards so as to face each other.

(12) For the sake of convenience, the left side in FIG. **4** will be referred to as a left side of the bracket **11**; the right side in FIG. **4** as a right side of the bracket **11**; the lower side in FIG. **4** as a front side of the bracket **11**; and the upper side in FIG. **4** as a rear surface side of the bracket **11**. Accordingly, FIG. **4** is a top view of the bracket **11**; FIG. **5** is a left side view of the bracket **11**; and FIG. **6** is a rear view of the bracket **11**. The bracket **11** is bilaterally symmetric about a vertical plane (hereinafter, referred to as “axial plane”) including an axial center A (axial line).

(13) As illustrated in FIGS. **2** and **5**, the extending portion **31** is provided with a first extending portion's hole **32** and a second extending portion's hole **33** which are vertically distanced from each other. Bolts, not shown, to be fastened to a knuckle **10** (see FIG. **1**) respectively extend through the first extending portion's hole **32** and the second extending portion's hole **33**. The bracket **11** includes a reinforcing portion **34** that is formed to extend from a left side surface **24** of the cylindrical portion **21** to the extending portion **31**. The reinforcing portion **34** is press-formed to expand in a radially outward direction of the cylindrical portion **21**.

(14) The reinforcing portion **34** includes a front-side (“right side” in FIG. **5**) end extending into between the first extending portion's hole **32** and the second extending portion's hole **33**. Positions (heights) of upper and lower ends of the reinforcing portion **34** in the axial direction (vertical direction) are substantially the same as axial positions of centers of the first and second extending portion's holes **32** and **33** of the extending portion **31**. Since the bracket **11** is bilaterally symmetric about the axial plane as already mentioned, explanations of the extending portion **41** will be omitted. Elements included in the extending portion **41** which correspond to elements in the extending portion **31** are illustrated in the drawings with reference numerals obtained by adding **10** to reference numerals for the corresponding elements of the extending portion **31**.

(15) Provided above the reinforcing portion **34** and at a higher level (higher position) than the first extending portion's hole **32** is a reinforcing portion **37** that is formed to extend from an upper end portion of the left side surface **24** of the cylindrical portion **21** to an upper end portion of the extending portion **31**. A lib **38** is formed at a lower end of the bracket **11**. The lib **38** is formed to extend in the radially outward direction of the cylindrical portion **21** from the left side surface **24** of the cylindrical portion **21** over the first end portion **22** (see FIG. **4**) to the extending portion **31**.

(16) As illustrated in FIGS. **1**, **5** and **6**, the bracket **11** includes a first hole **51** and a second hole **61** each having a fig-like shape which extend through the left side surface **24** and a right side surface **25** of the cylindrical portion **21**. The first hole **51** includes an arc-like portion **52** extending to have a substantially arc-like shape in a left-side-surface view along an outer periphery of the valve case **6** which protrudes radially outwards from the lateral wall of the cylinder **2**. The arc-like portion **52** is provided to contact the first end portion **22** (see FIG. **4**) of the left side surface **24** of the cylindrical portion **21** at a contact point B. The cylindrical portion **21** side (“left side” in FIG. **5**) of the reinforcing portion **34** is accordingly curved along the arc-like portion **52** of the first hole **51**. In

other words, the arc-like portion **52** (at least part of the first hole **51**) is formed to extend to the reinforcing portion **34** in a circumferential direction of the cylindrical portion **21** and formed between an upper end portion **35** and a lower end portion **36** of the reinforcing portion **34** in the axial direction of the cylindrical portion **21**.

(17) The arc-like portion **52** includes a lower edge portion **56** extending in a circumferential direction from a lower end **54** of the arc-like portion **52** to a rear surface side of the arc-like portion **52** ("left side" in FIG. 5) and an upper edge portion **55** extending obliquely upwards from the upper end **53** of the arc-like portion **52** through an R portion (reference sign omitted) to the rear surface side of the arc-like portion **52**. The upper edge portion **55** and the lower edge portion **56** are connected together through a longitudinal edge portion **57** extending in the axial direction (vertical direction). Since the bracket **11** is bilaterally symmetric about the axial plane as already mentioned, explanations of the second hole **61** will be omitted. Elements included in the second hole **61** which correspond to elements included in the first hole **51** are illustrated in the drawings with reference numerals obtained by adding **10** to reference numerals for the corresponding elements of the first hole **51**.

(18) The valve case **6** (protruding portion) protrudes from either one of the first and second holes **51** and **61**. In the embodiment, the valve case **6** protrudes from the first hole **51**. A bracket **7** is joined to the outer peripheral surface **2A** of the cylinder **2** which is exposed from the inside of the second hole **61**. Linkage components and the like are connected to the bracket **7**.

(19) As illustrated in FIG. 6, the bracket **11** comprises a bridge portion **71** that is provided at a center of a rear surface of the cylindrical portion **21**, that is, on the opposite side of the axial center **A** from the extending portions **31** and **41** (see FIG. 4). In other words, the bridge portion **71** is provided on a circumferentially opposite side of the cylindrical portion **21** from a center position between the first and second end portions **22** and **23** of the cylindrical portion **21**.

(20) The bridge portion **71** is a straight portion that is provided between the longitudinal edge portion **57** of the first hole **51** of the left side surface **24** of the cylindrical portion **21** and a longitudinal edge portion **67** of the second hole **61** of the right side surface **25** of the cylindrical portion **21** and extends in the axial direction ("up-and-down direction" in FIG. 6) at a substantially horizontal position to the axial direction of the cylindrical portion **21** with respect to an axial center position **C** (see FIG. 5) of the first extending portion's hole **32**. In other words, (at least part of) a left side surface **74** of the bridge portion **71** which is formed in the left side surface **24** of the cylindrical portion **21** is disposed at a substantially horizontal position to the axial direction of the cylindrical portion **21** with respect to the axial center position **C** of the first extending portion's hole **32**.

(21) The bridge portion **71** includes a decreased diameter portion **72** including a region (at least a part) in which width **H1** in the circumferential direction ("right-and-left direction" in FIG. 6) of the cylindrical portion **21** is decreased. The decreased diameter portion **72** is so provided that a position (height) thereof in the axial direction ("up-and-down direction" in FIG. 6) substantially coincides with a position of the contact point **B** (see FIG. 5) between the arc-like portion **52** and the first end portion **22**. In other words, the decreased diameter portion **72** is provided in the circumferential direction of the cylindrical portion **21** from the valve case **6** (protruding portion), that is, on a plane including an axial center (axial line) of the valve case **6**.

(22) The bridge portion **71** includes an increased diameter portion **73** including a region (at least a part) in which width **H2** in the circumferential direction of the cylindrical portion **21** is increased. The increased diameter portion **73** is so provided that a position (height) thereof in the axial direction ("up-and-down direction" in FIG. 6) coincides with height between a position of the upper end **53** (see FIG. 5) of the arc-like portion **52** and the center position **C** (see FIG. 5) of the first extending portion's hole **32**. As illustrated in FIGS. 2 and 3, an additional component **8**, such as a harness bracket, is joined (welded) to the increased diameter portion **73** of the bridge portion **71**.

(23) In conventional brackets, a hole for allowing a valve case (protruding portion) to extend therethrough is formed to extend from both side portions of a cylindrical portion to a rear portion of the cylindrical portion. The cylindrical portion is shaped by press-forming a hole in sheet material (flat plate) and then curving the sheet material in which the hole is formed into a tube. However, it is difficult to shape the sheet material (flat plate) that is widely open at the center into a tube with high accuracy. The accuracy of the tube-like portion relates to shock absorber attachment rigidity against the lateral force to the anteroposterior direction of a vehicle in a situation where the bracket is used to attach the shock absorber to the knuckle. The issue is therefore to improve attachment rigidity by improving the formability of the bracket.

(24) In contrast, according to the embodiment, the bracket **11** comprises the first hole **51** provided in the left side surface **24** of the cylindrical portion **21** and allowing the valve case **6** (protruding portion) to extend through the first hole **51**, the second hole **61** provided in the right side surface **25** of the cylindrical portion **21** and disposed apart from the first hole **51**, and the bridge portion **71** provided between the first hole **51** and the second hole **61** and disposed on the opposite side of the axial center A of the cylindrical portion **21** from the extending portions **31** and **41**.

(25) In the embodiment, the cylindrical portion **21** is shaped by press-forming the first hole **51** and the second hole **61** in the sheet material (flat plate) and then curving the sheet material into a tube, in which the bridge portion **71** is formed between the first hole **51** and the second hole **61**. This improves the formability of the bracket as compared to conventional brackets which are not provided with the bridge portion **71**, and therefore improves the accuracy (particularly, cylindricity) of the cylindrical portion **21**. This results in improvement in attachment rigidity of the shock absorber **1** against the lateral force to the anteroposterior direction of the vehicle in a situation where the bracket **11** is used to attach the shock absorber **1** to the knuckle **10**.

(26) In the embodiment, the reinforcing portion **34** and a reinforcing portion **44** are formed to extend from the side surfaces **24**, **25** of the cylindrical portion **21** to the extending portions **31**, **41**. The arc-like portion **52** and an arc-like portion **62** are part of the holes **51**, **61**. The arc-like portion **52** and an arc-like portion **62** are formed to extend to the reinforcing portions **34**, **44** in the circumferential direction of the cylindrical portion **21**. The arc-like portion **52** is formed between the upper end portion **35** and the lower end portion **36** of the reinforcing portion **34** in the axial direction of the cylindrical portion **21**. The arc-like portion **62** is formed between an upper end portion **45** and a lower end portion **46** of the reinforcing portion **44** in the axial direction of the cylindrical portion **21**. As a result of removal of regions (a region between the axial end portions **35**, **36** of the reinforcing portion **34** and **45**, **46** between the axial end portions of the reinforcing portion **44**) with minimal impact on strength, it is possible to reduce weight and secure strength at the same time.

(27) In the embodiment, since the bridge portion **71** is provided between the first hole **51** and the second hole **61**, it is possible to reduce the weight of the bracket **11** due to the first and second holes **51**, **61** and yet secure the attachment rigidity of the shock absorber **1** due to the bridge portion **71**.

(28) In the embodiment, the additional component **8** (fixing member), such as a harness bracket, is joined by welding to the increased diameter portion **73** of the bridge portion **71**. As compared to a case in which the additional component **8** is welded to the outer peripheral surface **2A** of the cylinder **2**, the embodiment is capable of suppressing embrittlement of material of the cylinder **2** which is caused by welding heat effect and securing the attachment rigidity of the shock absorber **1**.

(29) In the embodiment, the decreased diameter portion **72** is provided in the bridge portion **71** at the position in the circumferential direction of the cylindrical portion **21** from the valve case **6**. When the valve case **6** is joined (welded) to the cylinder **2**, therefore, a gap can be secured between the valve case **6** and the bridge portion **71**.

(30) The invention is not limited to the embodiment and may be modified in various ways. For

example, the embodiment is intended to facilitate the understanding of the invention and does not necessarily have to comprise all the configurations explained above. The configuration of each embodiment may be partially replaced with another configuration or incorporated with another configuration. It is also possible to incorporate, omit or replace a part of the configuration of one of the embodiments into, from or with the configuration of another one of the embodiments.

(31) The present application claims priority under Japanese Patent Application No. 2022-15777 filed on Feb. 3, 2022. The entire disclosure of Japanese Patent Application No. 2022-15777 filed on Feb. 3, 2022 including the description, claims, drawings and abstract, is incorporated herein by reference in its entirety.

REFERENCE SIGN LIST

(32) **1** Shock absorber; **2** Cylinder; **6** Valve case (protruding portion); **11** Bracket; **21** Cylindrical portion; **22** First end portion; **23** Second end portion; **31, 41** Extending portion; **51** First hole; **61** Second hole; **71** Bridge portion

Claims

1. A shock absorber comprising: a tube-like cylinder; a protruding portion that is so provided as to protrude in a radially outward direction of the cylinder; and a bracket that is provided at an outer periphery of the cylinder, the bracket comprising: a cylindrical portion including a first end portion and a second end portion in a circumferential direction of the cylindrical portion, the cylindrical portion having a cross-section in an axial direction of the cylindrical portion, the cross-section in the axial direction being formed into a C-shape, the cylindrical portion covering the outer periphery of the cylinder; a pair of extending portions respectively extending from the first end portion and the second end portion of the cylindrical portion in the radially outward direction of the cylindrical portion so as to face each other; a first hole that is provided in the cylindrical portion and allows the protruding portion to extend through the first hole in the radially outward direction of the cylindrical portion; a second hole that is provided in the cylindrical portion and disposed apart from the first hole; and a bridge portion that is formed between the first hole and the second hole and provided on an opposite side of an axial center of the cylindrical portion from the pair of extending portions, wherein the bridge portion includes a decreased width portion that is at least partially decreased in width, the decreased width portion being provided in the circumferential direction of the cylindrical portion from the protruding portion.

2. The shock absorber according to claim 1, wherein a fixing member is provided at a radially outer side of the bridge portion.

3. The shock absorber according to claim 2, wherein the fixing member is provided to an increased width portion of the bridge portion which is at least partially increased in width.

4. The shock absorber according to claim 1, including: reinforcing portions that are provided at a radially outer side of the cylindrical portion and formed to extend from the cylindrical portion to the respective extending portions, wherein the first hole or the second hole is at least partially formed to extend to the corresponding reinforcing portion in the circumferential direction of the cylindrical portion and formed between end portions of the corresponding reinforcing portion which are located in an axial direction of the cylindrical portion.

5. The shock absorber according to claim 1, wherein a first extending portion's hole and a second extending portion's hole are formed in the pair of extending portions, the first extending portion's hole allowing a rod-like member to extend through the first extending portion's hole, and the second extending portion's hole being disposed apart from the first extending portion's hole and allowing a rod-like member to extend through the second extending portion's hole, and wherein at least a part of the bridge portion is disposed at a substantially horizontal position to the axial direction of the cylindrical portion with respect to an axial center position of the first extending portion's hole.

6. The shock absorber according to claim 5, wherein the bridge portion is a straight portion that extends in the axial direction of the cylindrical portion at the substantially horizontal position to the axial direction of the cylindrical portion with respect to the axial center position of the first extending portion's hole.
7. The shock absorber according to claim 1, wherein the bridge portion is formed between the first hole and the second hole and provided on a circumferentially opposite side of the cylindrical portion from a center position of the cylindrical portion which is located between the first end portion and the second end portion.
8. A shock absorber comprising: a tube-like cylinder; a protruding portion that is so provided as to protrude in a radially outward direction of the cylinder; and a bracket that is provided at an outer periphery of the cylinder, the bracket comprising: a cylindrical portion including a first end portion and a second end portion in a circumferential direction of the cylindrical portion, the cylindrical portion having a cross-section in an axial direction of the cylindrical portion, the cross-section in the axial direction being formed into a C-shape, the cylindrical portion covering the outer periphery of the cylinder; a pair of extending portions respectively extending from the first end portion and the second end portion of the cylindrical portion in the radially outward direction of the cylindrical portion so as to face each other; a first hole that is provided in the cylindrical portion and allows the protruding portion to extend through the first hole in the radially outward direction of the cylindrical portion; a second hole that is provided in the cylindrical portion and disposed apart from the first hole; and a bridge portion that is formed between the first hole and the second hole and provided on an opposite side of an axial center of the cylindrical portion from the pair of extending portions, the shock absorber further including: reinforcing portions that are provided at a radially outer side of the cylindrical portion and formed to extend from the cylindrical portion to the respective extending portions, wherein the first hole or the second hole is at least partially formed to extend to the corresponding reinforcing portion in the circumferential direction of the cylindrical portion and formed between end portions of the corresponding reinforcing portion which are located in an axial direction of the cylindrical portion.
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